Econ 131
Spring 2026

Midterm 2 Draft

Monday, April 6, 2026

Student Name:

Student ID:

Exam Instructions

- **This exam has three questions. Question 1 and Question 3 are worth 60 points each, and Question 2 is worth 30 points (150 points total). The total exam time is 75 minutes.**
- **Write answers using pens (we recommend black or blue ink), and write only in the space provided for each answer so your solutions scan properly.**
- **Do not write your solutions on pages that say "Do not write on this page"**. Answers written on these pages will not be graded.
- **Show your work**. Credit will only be awarded on the basis of what is written on the exam.
- **Affirm the academic honesty pledge below**. For those writing on a non-printed copy, please write "Academic Honesty Pledge as on exam" and sign your name. **If you do not affirm this pledge, your exam will be marked invalid.**

0. ACADEMIC HONESTY PLEDGE

I confirm that I have abided by all academic honesty rules for UC Berkeley and Economics 131. I confirm that I did not see this exam before my official exam start time. I confirm that I have not shared and will not share this exam with anyone else.

Signature: ____________________

1. True/False/Uncertain Questions (60 points total)

You can start your answer "True", "False", "Uncertain", "Partially true", or "Partially false", etc., but your choice of the label does not matter much. What matters is your explanation.

(a) Suppose that a tax reform cuts the corporate tax rate and makes investment immediately deductible, rather than deductible over time. The tax reform will increase investment.

PARTIALLY TRUE. Moving from delayed depreciation deductions to immediate deduction (expensing) raises the present value of write-offs and lowers the cost of capital for new marginal investment, so that part of the reform pushes toward more investment. But in the simple full-expensing logic emphasized in class, the corporate tax rate itself does not necessarily affect the normal return to a new marginal investment once investment is immediately deductible. So the clearest reason to expect more investment here is the move to immediate expensing, not the rate cut by itself.

(b) If higher-skill individuals systematically earn higher returns on wealth than lower-skill individuals, the baseline Atkinson-Stiglitz case for a zero capital income tax becomes weaker.

TRUE. Atkinson-Stiglitz says that with an optimal nonlinear labor tax, capital taxes are unnecessary because skill differences are fully captured by labor income. But if high-skill individuals also earn higher returns on capital (e.g., through access to better investments or ability to bear risk), then capital income reveals additional information about ability beyond what labor income shows. Taxing capital income then improves redistribution, weakening the zero-capital-tax result.

(c) Suppose a town's new data center uses a lot of electricity, which makes electricity more expensive for the town's families. This is an example of a market failure.

FALSE. A higher electricity price caused by greater demand is a *pecuniary* externality, not a technological externality. The price increase works through the market and reflects real scarcity. By itself, that does not imply market failure. It would become a market-failure story only if the data center also created some unpriced external harm, such as pollution or strain on infrastructure not reflected in prices.

(d) When pollution abatement costs are uncertain and when the marginal damage from pollution does not depend on the pollution level, tradable permits can be preferable to a corrective tax.

FALSE. If marginal damages are flat or do not depend on the pollution level, then getting the exact quantity right is less important. In that case, Weitzman's prices-versus-quantities logic points toward a **tax** being preferable, because a tax stabilizes the marginal cost of abatement while permits can create inefficient quantity rigidity when abatement costs are uncertain.

(e) For private goods, social efficiency requires summing marginal rates of substitution across consumers in the same way as for public goods.

FALSE. For private goods (rival), efficiency requires each individual's MRS equal the marginal cost: $MRS_1 = MRS_2 = MC$. For public goods (non-rival), the Samuelson condition requires *summing* MRS across consumers: $MRS_1 + MRS_2 = MC$, because all consumers benefit simultaneously from the same unit. The two conditions are fundamentally different.

(f) Warm-glow motives can imply less than one-for-one crowd-out of private giving by government provision.

TRUE. In a pure altruism model, government provision perfectly crowds out private giving one-for-one because individuals care only about total provision. With warm-glow motives, people also value the act of giving itself, so they may continue to donate even when the government provides more. That implies crowd-out can be less than complete.

(g) Suppose Berkeley residents use Oakland's public playgrounds. The Tiebout model says that Oakland and Berkeley public playgrounds will nevertheless be efficiently provided since people "vote with their feet".

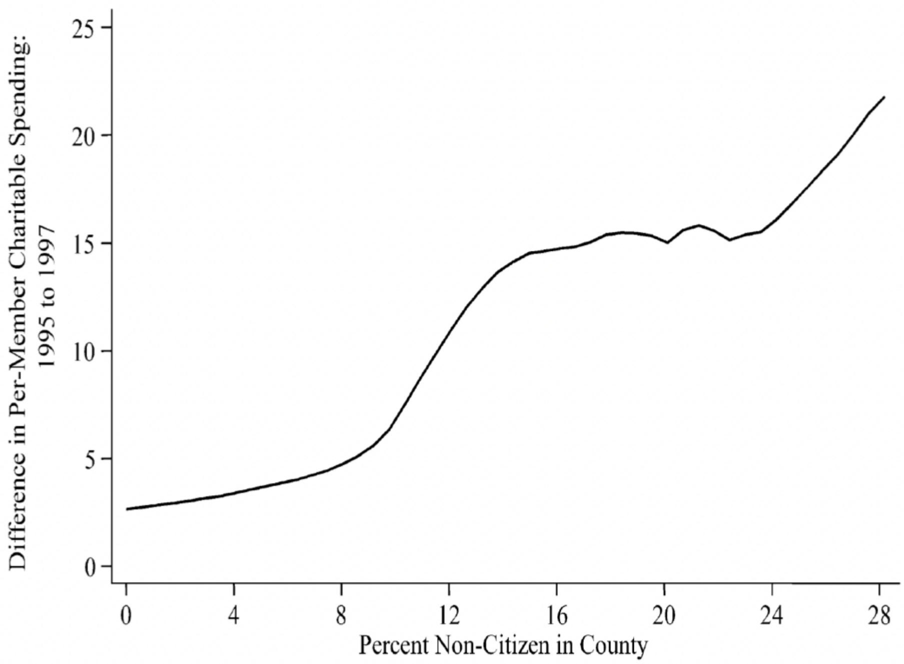
FALSE. Tiebout efficiency requires **no spillovers** across jurisdictions. If Berkeley residents benefit from Oakland's playgrounds, then Oakland does not internalize all of the benefits from its own spending. That creates a positive externality across towns and tends to produce under-provision relative to the social optimum. "Voting with your feet" does not solve this spillover problem by itself.

(h) People know the quality of local public goods.

FALSE / UNCERTAIN. Perfect information about local taxes and local public-good quality is a strong **assumption** in the idealized Tiebout model, not a general fact about the real world. Ferraz-Finan (2008) shows that when Brazilian municipal audits publicly revealed corruption, voters became less likely to reelect corrupt incumbent mayors. That evidence suggests voters respond to new information about local government quality, which in turn suggests they do *not* already observe that quality perfectly.

2. Empirical Application: Hungerman (2005) and Crowd-Out of Church Charity (30 points total)

In 1996, welfare reform reduced or eliminated some transfers for non-citizens. Hungerman (2005) studies whether church-provided charitable spending rose more in places more exposed to this reform. The figure below plots the change in per-member charitable spending from 1995 to 1997 against the percent non-citizen in the county.



Source: Hungerman 2005

(a) Explain in words the difference-in-differences style conclusion this figure is trying to suggest. What broad pattern in the figure would be consistent with the idea that counties more exposed to the welfare cuts experienced larger increases in church charitable spending? *(7.5 points)*

Solution:

The figure is trying to suggest a difference-in-differences style conclusion using **treatment intensity**. Counties with more non-citizens were more exposed to the 1996 welfare reform, so if the welfare cuts caused churches to step in more, then the more-exposed counties should experience larger post-reform increases in church charitable spending.

The broad pattern consistent with that story is an **upward-sloping relationship**: counties with higher non-citizen shares should show larger increases in per-member church charitable spending from 1995 to 1997.

(b) Based on this figure alone, do you find the causal interpretation convincing? Answer yes or no and explain briefly. What additional evidence or alternative figure would you want to see to make the claim more convincing? *(7.5 points)*

Solution:

No, **not by itself**. This figure is only a cross-sectional plot of changes against exposure, so it does not show whether high-exposure and low-exposure counties were already on different trends before the reform or whether some other county characteristic is driving both non-citizen share and church spending growth.

To make the causal claim more convincing, I would want a **difference-in-differences/event-study figure**: plot per-member church charitable spending over time for high-exposure versus low-exposure counties, with a vertical line at 1996. That would let us assess pre-trends and see whether the groups diverged only after the reform.

(c) Let $Treat_c = 1$ denote the counties with a high non-citizen share of the population, and let $Treat_c = 0$ denote otherwise. Let $Post_t = 1$ denote years after the reform and $Post_t = 0$ denote years before the reform. Let Y_{ct} denote church charitable spending in county c and year t. Suppose you estimate the regression

$$Y_{ct} = \alpha + \beta Treat_c + \gamma Post_t + \delta(Treat_c \times Post_t) + \varepsilon_{ct}.$$

Using the coefficients $\alpha, \beta, \gamma, \delta$ from this regression, fill in the four cells in the following table. Each cell equals a mean. For example, the control/pre cell equals the mean value of Y_{ct} in the control counties before the reform. *(7.5 points)*

Solution:

Plug in the values of $Treat_c$ and $Post_t$ for each cell:

	Pre ($Post_t = 0$)	Post ($Post_t = 1$)
Control ($Treat_c = 0$)	α	$\alpha + \gamma$
Treatment ($Treat_c = 1$)	$\alpha + \beta$	$\alpha + \beta + \gamma + \delta$

(d) Which regression coefficient or coefficients equal the difference-in-differences estimate? Briefly interpret what a positive value of that coefficient or coefficients would mean in this context. *(7.5 points)*

Solution:

The difference-in-differences estimate equals the coefficient $\boldsymbol{\delta}$ on the interaction term $Treat_c \times Post_t$.

A positive δ would mean that church charitable spending increased *more* in treatment counties (those more exposed to the welfare cuts) than in control counties after the reform. That is evidence of **crowd-in**: places hit harder by the reduction in government welfare saw a larger increase in church-provided charitable spending.

3. Negative Income Tax vs. Earned Income Tax Credit (60 points total)

A city has 1,000 working-age residents. Each resident decides whether to work or not. A worker earns pre-tax labor income $z > 0$; a non-worker earns $z = 0$. Initially there are **no taxes or transfers**.

(a) The city introduces a **Negative Income Tax (NIT)**:

- Every resident receives a **guaranteed income of $20** (even with zero earnings).
- The guarantee is **phased out at 50 cents per dollar** of earnings.
- There are **no further taxes** above the point where the guarantee is fully phased out.

At what level of earnings z does the guarantee fully phase out? Write after-tax-and-transfer income $c(z) = z - T(z)$ for each segment, and draw the budget constraint with pre-tax earnings z on the x-axis and c on the y-axis. Label the y-intercept, any kink points (with coordinates), and the slope of each segment. *(12 points)*

Solution:

The guarantee of $20 is reduced by $0.50 for each dollar earned, so it is fully phased out when $0.50 \times z = 20$, i.e., at $\mathbf{z}^* = \mathbf{40}$.

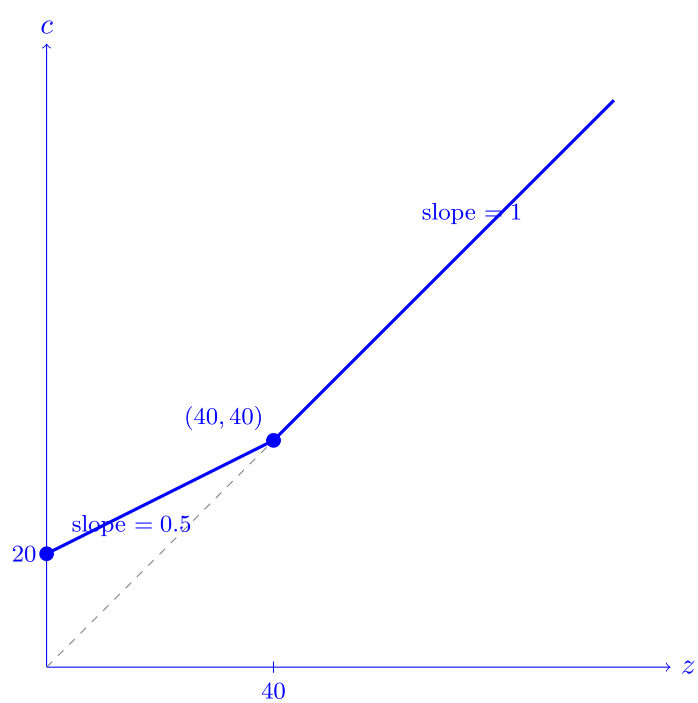
Segment 1 ($0 \leq z \leq 40$): The resident receives the grant minus the phase-out:

$$c(z) = z + (20 - 0.5z) = 20 + 0.5z, \qquad \text{slope} = 0.5$$

Segment 2 ($z > 40$): The grant is fully phased out and there are no taxes:

$$c(z) = z, \qquad \text{slope} = 1$$

Key points: y-intercept (0, 20); kink at (40, 40). The budget constraint is continuous at the kink since $20 + 0.5(40) = 40$.



(b) Using the NIT from part (a), compute the **participation tax rate**

$$\tau_p \;=\; \frac{T(z) - T(0)}{z}$$

for a worker earning $z = 20$ and for a worker earning $z = 80$. Which faces the higher participation tax rate? In one sentence, explain intuitively why a system with a phased-out guaranteed income creates especially high participation tax rates for low earners. *(12 points)*

Solution:

First note that $T(0) = -20$ (a non-worker receives a \$20 grant, so net tax is -20).

Worker at $z = 20$: In the phase-out region. $c(20) = 20 + 0.5(20) = 30$, so $T(20) = 20 - 30 = -10$.

$$\tau_p = \frac{T(20) - T(0)}{20} = \frac{-10 - (-20)}{20} = \frac{10}{20} = \mathbf{50}\%$$

Worker at $z = 80$: Above the phase-out. $c(80) = 80$, so $T(80) = 0$.

$$\tau_p = \frac{T(80) - T(0)}{80} = \frac{0 - (-20)}{80} = \frac{20}{80} = \mathbf{25}\%$$

The **low earner faces the higher participation tax rate** (50% vs. 25%). Intuitively, when a worker starts working she gives up the \$20 grant; this fixed "cost" of entering the labor force represents a much larger fraction of a low earner's income than of a high earner's income.

(c) Before the NIT was introduced, there were 700 workers and 300 non-workers. After the NIT is introduced:

- 100 workers who had been earning $z = 20$ each **stop working**, discouraged by the high participation tax rate.
- 500 workers who earn $z = 80$ each continue working.
- The remaining 100 workers who earn $z = 20$ each also continue working.
- The 300 original non-workers remain out of work.

Compute the government's **total cost** of the NIT (total grants paid out, net of any phase-out clawback). *(12 points)*

Solution:

After the behavioral response there are 400 non-workers and 600 workers.

- **400 non-workers** ($z = 0$): Each receives the full grant of \$20. Cost $= 400 \times 20 =$ \$8,000.
- **100 workers at** $z = 20$: Grant received $= 20 - 0.5(20) =$ \$10 each. Cost $= 100 \times 10 =$ \$1,000.
- **500 workers at** $z = 80$: Since $80 > 40$, the grant is fully phased out. Cost $= 500 \times 0 =$ \$0.

$$\boxed{\textbf{Total NIT cost} = 8{,}000 + 1{,}000 + 0 = \$9{,}000}$$

(d) The city replaces the NIT with an **Earned Income Tax Credit (EITC)**:

- **No payment to non-workers**: $c(0) = 0$.
- **Phase-in** ($z = 0$ to \$20): workers receive a subsidy of \$0.50 per dollar earned.
- **Plateau** ($z =$ \$20 to \$40): workers receive a flat credit of \$10.
- **Phase-out** (z above \$40): the credit is reduced by \$0.25 per dollar of earnings above \$40.
- No other taxes.

At what earnings level is the credit fully phased out? Draw the budget constraint (same axes as part (a)). Label the y-intercept, all kink points (with coordinates), and the slope of each segment. *(12 points)*

Solution:

The maximum credit is $0.50 \times 20 = \$10$, reached at $z = 20$. It stays at \$10 through the plateau. In the phase-out, it falls by \$0.25 per dollar above \$40, so it is fully gone when $0.25(z - 40) = 10$, i.e., at $\mathbf{z = 80}$.

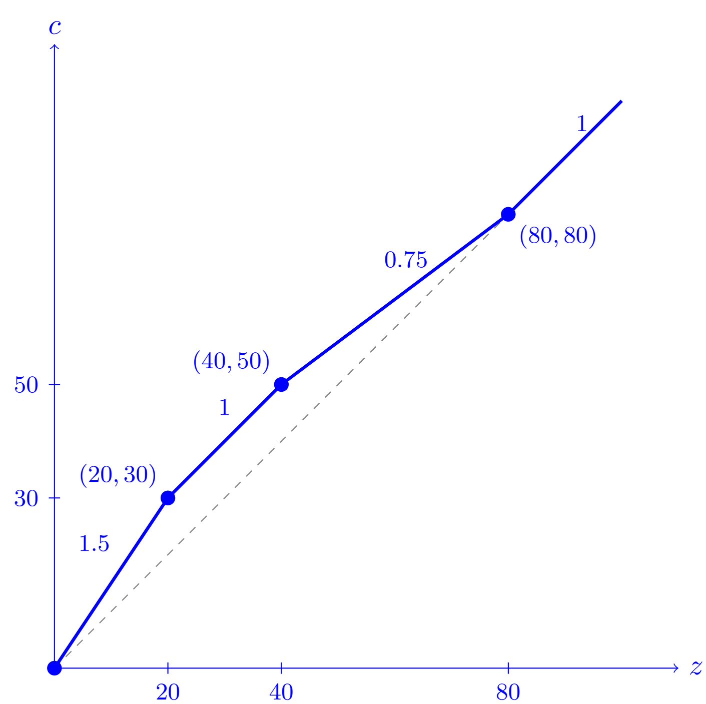
Segment 1 — Phase-in ($0 \leq z \leq 20$): $c(z) = z + 0.5z = 1.5z$, slope $= 1.5$

Segment 2 — Plateau ($20 \leq z \leq 40$): $c(z) = z + 10$, slope $= 1$

Segment 3 — Phase-out ($40 \leq z \leq 80$): $c(z) = z + 10 - 0.25(z - 40) = 0.75z + 20$, slope $= 0.75$

Segment 4 ($z > 80$): $c(z) = z$, slope $= 1$

Kink points: (0, 0), (20, 30), (40, 50), (80, 80). All segments continuous.



(e) Motivated by the EITC structure in part (d), the government considers a modest reform to the NIT from part (c): it **reduces the phase-out rate from 50% to 25%** on the first \$20 of earnings. Above \$20, the phase-out rate remains 50%. All else stays the same (the guaranteed income is still \$20). *(12 points total)*

(i) What is the participation tax rate at $z = 20$ under this reformed NIT? Compare it to the original NIT participation tax rate from part (b). *(4 points)*

Solution:

Under the reformed NIT, the phase-out on the first \$20 is only 25%, so the grant received at $z = 20$ is $20 - 0.25 \times 20 = \$15$. After-tax income: $c(20) = 20 + 15 = 35$, so $T(20) = 20 - 35 = -15$. Non-workers still get \$20, so $T(0) = -20$.

$$\tau_p = \frac{T(20) - T(0)}{20} = \frac{-15 - (-20)}{20} = \frac{5}{20} = \mathbf{25}\%$$

This is **half** the original NIT participation tax rate of 50% from part (b). The reform makes it substantially more attractive to enter the labor force at $z = 20$.

(ii) The reform has a **mechanical cost**: the 100 workers at $z = 20$ who were already working now receive a larger net transfer. How much extra does each one cost the government? What is the total mechanical cost? *(4 points)*

Solution:

Under the original NIT, each $z = 20$ worker received a grant of $20 - 0.5(20) = \$10$.

Under the reformed NIT, each receives $20 - 0.25(20) = \$15$.

Extra cost per worker $= 15 - 10 = \$5$.

$$\boxed{\text{Total mechanical cost} = 100 \times \$5 = \$500}$$

(iii) Suppose that among the 200 residents with potential earnings $z = 20$, the **extensive-margin elasticity of the number who work** with respect to the net-of-tax rate $1 - \tau_p$ is

$$\varepsilon = \frac{\%\Delta\text{workers}}{\%\Delta(1 - \tau_p)} = 2.$$

Using the original NIT in part (c) as the baseline, compute how many of the 100 workers who had dropped out would re-enter when the participation tax rate falls from 50% to 25%. Then compute the **fiscal saving** from each returning worker: compare what they cost as a non-worker versus what they cost as a worker under the reformed NIT. What is the total extensive-margin saving? Does it cover the mechanical cost from part (ii)? Explain in 1–2 sentences why this type of reform can be a "free lunch." *(4 points)*

Solution:

Under the original NIT, the participation tax rate at $z = 20$ is 50%, so the net-of-tax rate is $1 - \tau_p = 0.50$.

Under the reformed NIT, the participation tax rate at $z = 20$ is 25%, so the net-of-tax rate is $1 - \tau_p = 0.75$.

Thus the percent increase in the net-of-tax rate is

$$\frac{0.75 - 0.50}{0.50} = 0.50 = 50\%.$$

With elasticity $\varepsilon = 2$, the percent increase in the number of $z = 20$ residents who work is

$$\%\Delta\text{workers} = \varepsilon \times \%\Delta(1 - \tau_p) = 2 \times 50\% = 100\%.$$

Originally, 100 of the 200 residents with potential earnings $z = 20$ were working, so a 100% increase means the number working rises from 100 to 200. Therefore, **100 workers re-enter the labor force**.

As a non-worker, each returning resident costs the government \$20 (the full grant).

As a worker at $z = 20$ under the reformed NIT, each costs \$15 (the reduced grant).

Fiscal saving per returning worker = $20 - 15 = \$5$.

$$\boxed{\text{Total extensive-margin saving} = 100 \times \$5 = \$500}$$

The extensive-margin saving (\$500) exactly covers the mechanical cost (\$500), so the reform **breaks even**—and if even a few additional workers return, it becomes a net fiscal gain.

This is the "free lunch" logic: reducing the participation tax rate costs money on workers who would have worked anyway (mechanical cost), but it saves money by drawing non-workers into the labor force, where they receive a smaller net transfer. When extensive-margin responses are large enough, the savings exceed the cost, making the reform self-financing.

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